

LVT overview

The Land-surface Verification Toolkit

David M. Mocko
SAIC at NASA/GSFC

LVT: Motivation

Entekhabi et al., *BAMS* (1999)



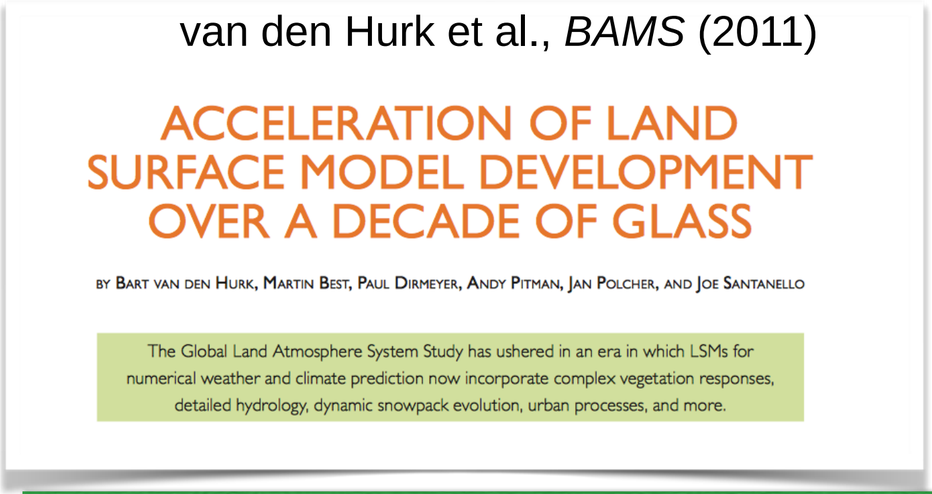
An Agenda for Land Surface Hydrology Research and a Call for the Second International Hydrological Decade

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ABSTRACT

Hydrologic research at the interface between the atmosphere and land surface is undergoing a dramatic change in focus, driven by new societal priorities, emerging technologies, and better understanding of the earth system. In this paper an agenda for land surface hydrology research is proposed in order to open the debate for more comprehensive

van den Hurk et al., *BAMS* (2011)



ACCELERATION OF LAND SURFACE MODEL DEVELOPMENT OVER A DECADE OF GLASS

BY BART VAN DEN HURK, MARTIN BEST, PAUL DIRMEYER, ANDY PITMAN, JAN POLCHER, AND JOE SANTANELLO

The Global Land Atmosphere System Study has ushered in an era in which LSMs for numerical weather and climate prediction now incorporate complex vegetation responses, detailed hydrology, dynamic snowpack evolution, urban processes, and more.

Need formal evaluation procedures to improve the "observability" of LSM processes

Need a general benchmarking framework capable of capturing useful modes of variability of LSMs through a range of performance metrics is necessary for further advancing the performance and predictability of models

- Quantitative measures of fidelity of model simulations are essential for improving the usage and acceptability of model forecasts for real-world applications
- Characterization of accuracy and uncertainty in model predictions - to be used as a benchmark for future model enhancements

LVT in the Model-Data-Fusion (MDF) paradigm

Williams et al., *Biogeosciences* (2009)

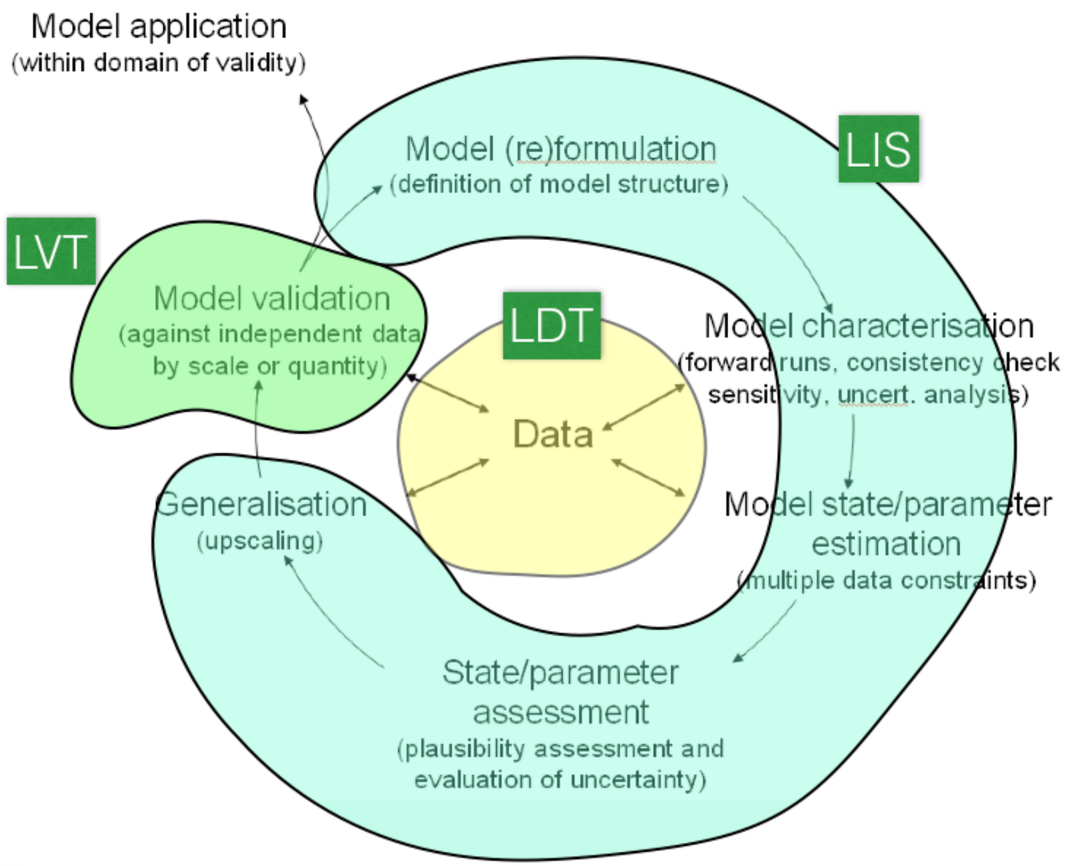


Fig. 1. The multi-stage process for model-data fusion: a conceptual diagram showing the main steps (and the iterative nature of these steps) involved in a comprehensive data-model fusion.

- MDF – the paradigm for combining information from models and data
- Use the information from data to help the formulation, characterization, and evaluation of models in a structured manner
- MDF and Benchmarking are two of the core themes of the GEWEX GLASS community
- A comprehensive evaluation and benchmarking framework is essential for enabling the MDF concept

LVT: Definitions

LVT functions both as a verification and as a benchmarking environment

The Plumbing of Land Surface Models: Benchmarking Model Performance

M. J. Best,[a] G. Abramowitz,[b] H. R. Johnson,[a] A. J. Pitman,[b] G. Balsamo,[c] A. Boone,[d] M. Cuntz,[e] B. Decharme,[d] P. A. Dirmeyer,[f] J. Dong,[g] M. Ek,[g] Z. Guo,[f] V. Haverd,[h] B. J. J. van den Hurk,[i] G. S. Nearing,[j] B. Pak,[k] C. Peters-Lidard,[j] J. A. Santanello Jr.,[j] L. Stevens,[k] and N. Vuichard[l]

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ABSTRACT

The Protocol for the Analysis of Land Surface Models (PALS) Land Surface Model Benchmarking Evaluation Project (PLUMBER) was designed to be a land surface model (LSM) benchmarking intercomparison. Unlike the traditional methods of LSM evaluation or comparison, benchmarking uses a fundamentally different approach in that it sets expectations of performance in a range of metrics a priori—before model simulations are performed. This can lead to very different conclusions about LSM performance. For this study, both simple

ver·i·fi·ca·tion
/ˌverəfəˈkāSH(ə)n/

noun

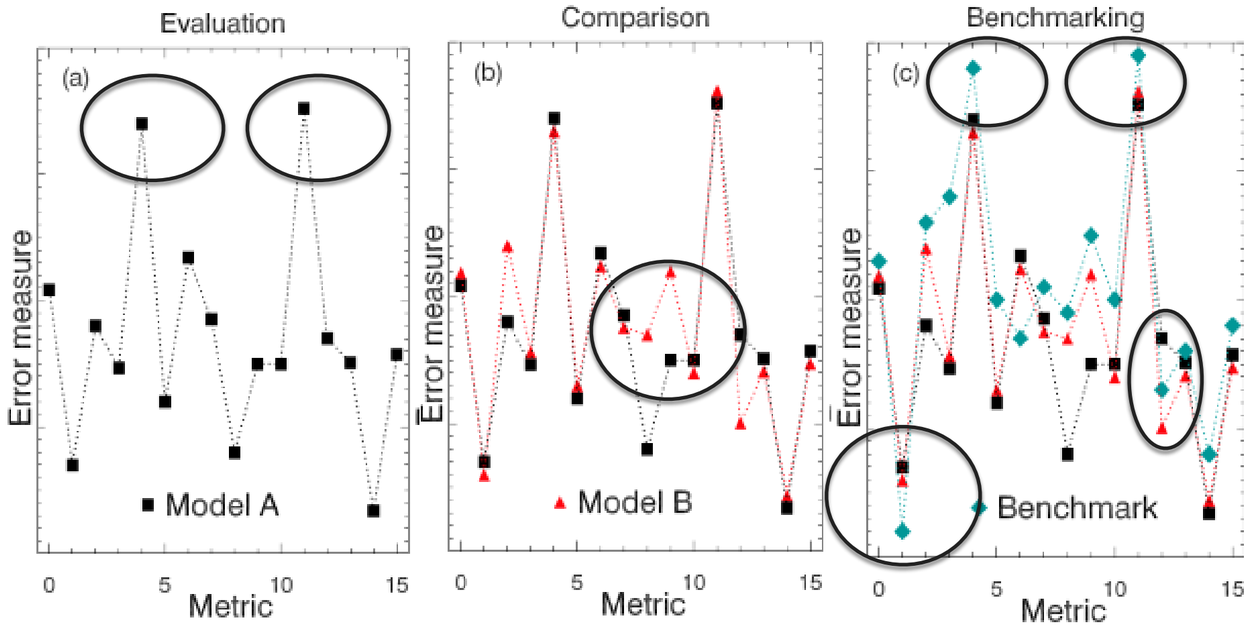
the process of establishing the truth, accuracy, or validity of something.
"the verification of official documents"
synonyms: confirmation, substantiation, proof, corroboration, support, attestation, validation, authentication, endorsement
"they

bench·mark
/ˈben(t)SHmärk/

verb
gerund or present participle: **benchmarking**

evaluate or check (something) by comparison with a standard.
"we are **benchmarking** our performance **against** external criteria"

source: Best et al. (2015)



- **Evaluation** - model outputs are compared to observations to derive an error measure
- **Comparison** - model is not just compared to observations, but also to other models
- **Benchmarking** - performance expectation defined a priori

Model Intercomparisons Projects (MIPs)



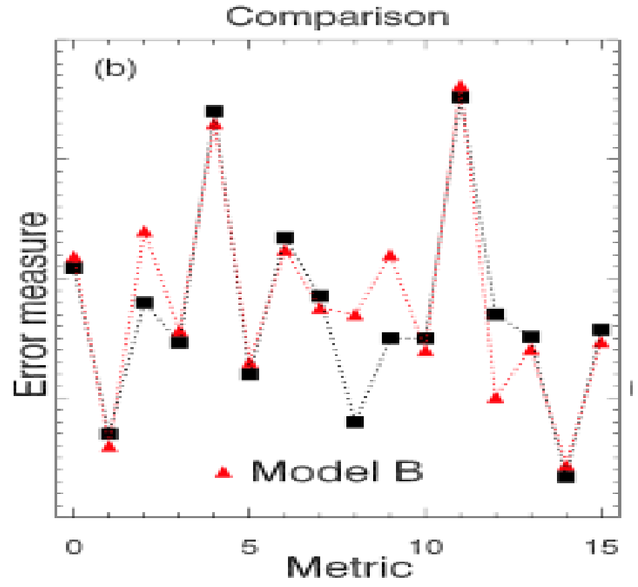
- Identifies metrics for which one model performs better than another, or where errors in multiple models are systematic
- Indicates where performance improvements are possible/not possible relative to other models
- Too much reliance on model comparisons - models may end up being developed too similar to each other

ALMIP AMMA Land Surface Model Intercomparison Project

ILAMB
THE INTERNATIONAL LAND MODEL BENCHMARKING PROJECT

C-LAMP is the
Carbon-Land Model
intercomparison Project

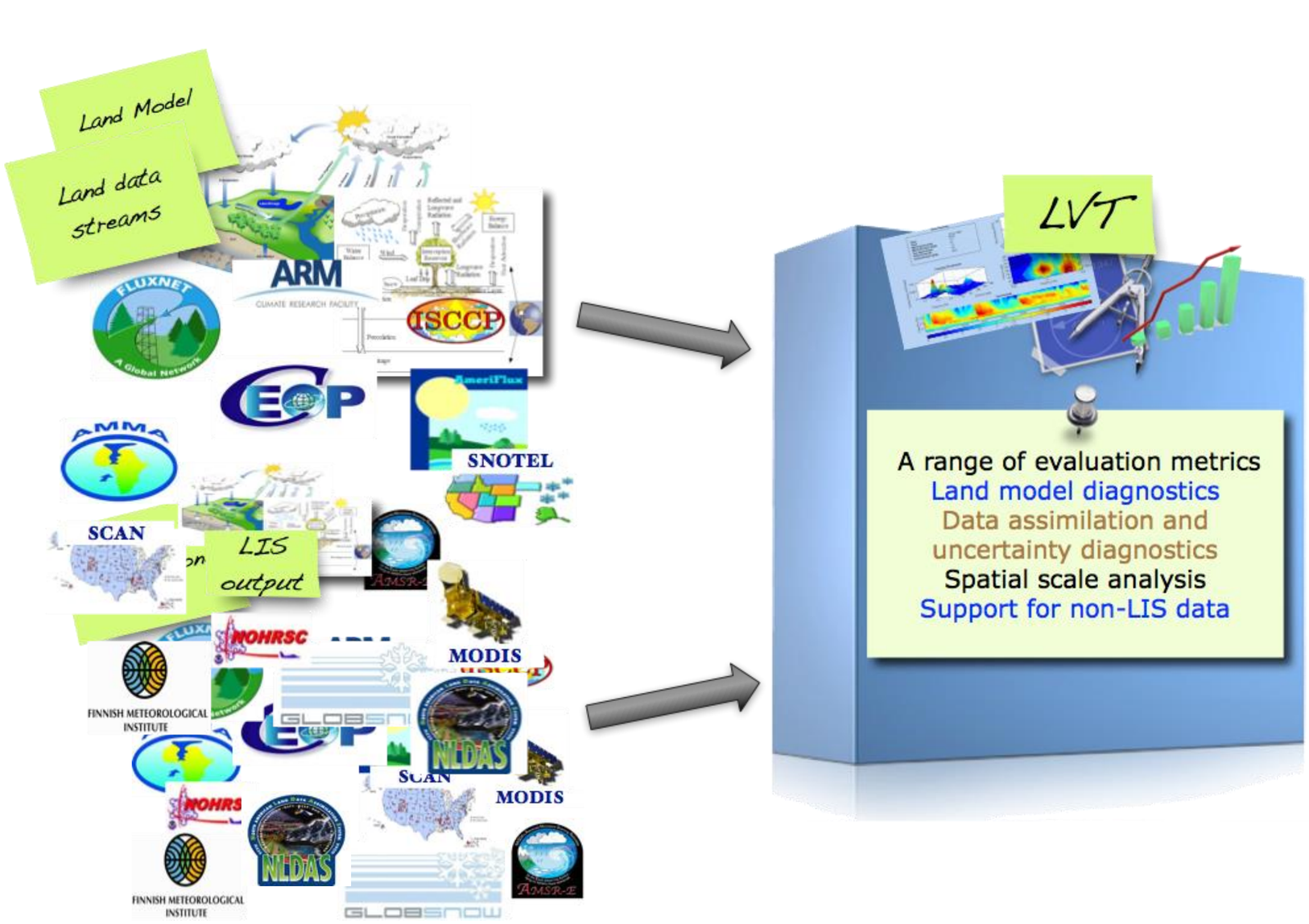
Carbon-Land Model Interc



Features of LVT

- LVT can be used in many different configurations/domains (e.g., NCEP, NOHRSC, CRREL, NRL, NLDAS, GLDAS, FLDAS, MSFC, NU-WRF, ICBA, …)
- LVT can process LIS outputs in many different formats (grib, NetCDF, binary), different resolutions, map projections, modes (tile, grid, ensembles)
- LVT was originally designed to have a framework that allows the comparison of LIS output against other observational/reference datasets for evaluation
- Includes support for a range of in-situ, remote sensing, and model/reanalysis products
- Supports the analysis of outputs from various LIS subsystems (LIS-DA, LIS-OPT/UE)
- Includes the capability to generate end-user oriented hydrological products (drought/flood percentiles, indicators)
- Handles reference datasets in their “native” format, so users can download these data directly from their archives and use them within LVT

LVT: Structure

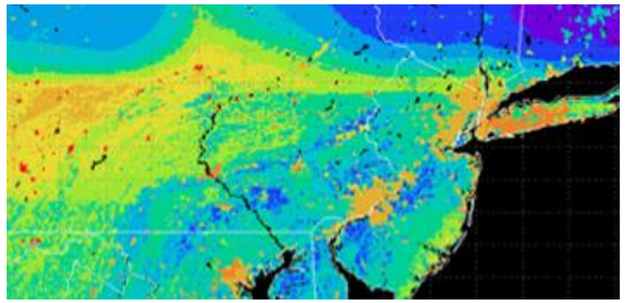


- LVT can handle any two land-relevant datasets (need not be a LIS output)
- Includes benchmarking capabilities as well as evaluation
- The supported datasets in LVT can be used to develop benchmarks using simple (regression) to more complex methods

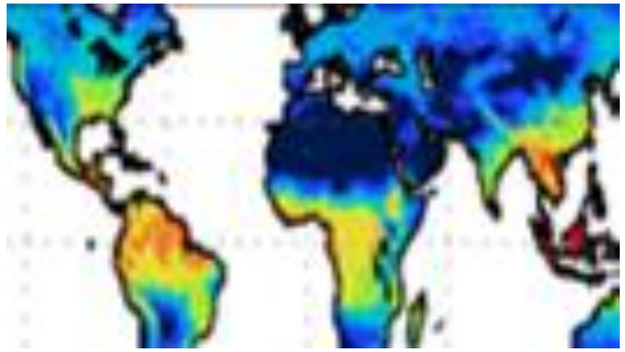
LVT: General capabilities

- Reconciles the differences in spatial and temporal resolutions between the two datastreams being compared, by bringing them to a common (user-specified) space and time domain

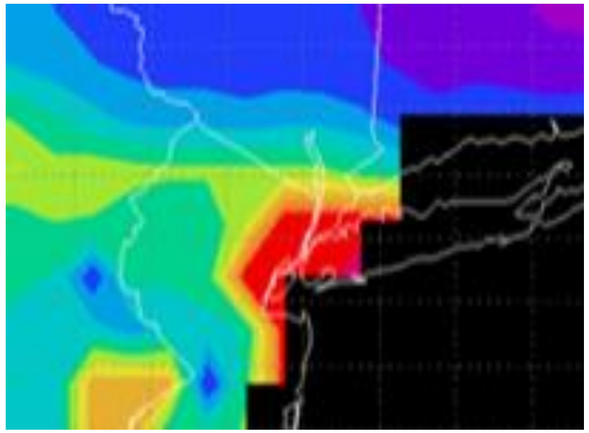
Datastream 1



Datastream 2

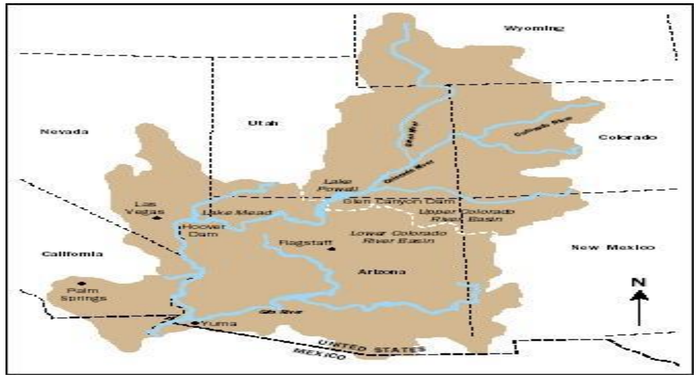


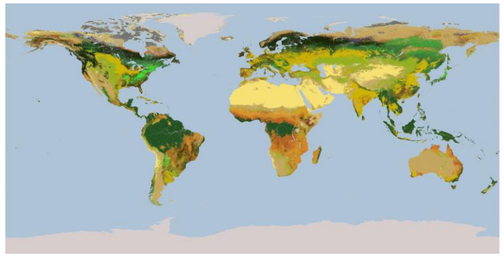
Analysis domain, resolution
(user specified)



LVT: General capabilities

- Emphasis on supporting datasets natively, as much as possible - Users can download the data directly and employ them in LVT
  - e.g., ARM-CART (NetCDF), AGRMET (Grib), SCAN in-situ (ASCII) …
  - A reader/processor needs to be built for each dataset
- Many options for masking/stratification of metrics:
  - Data count based mask
  - External static mask
  - External time varying mask
  - Variable-based stratification (e.g., day-night stratification using SWdown)
  - External data based stratification (e.g., landcover, soils, elevation)





RMSE for Evergreen Needleleaf Forest

RMSE for Evergreen Broadleaf Forest

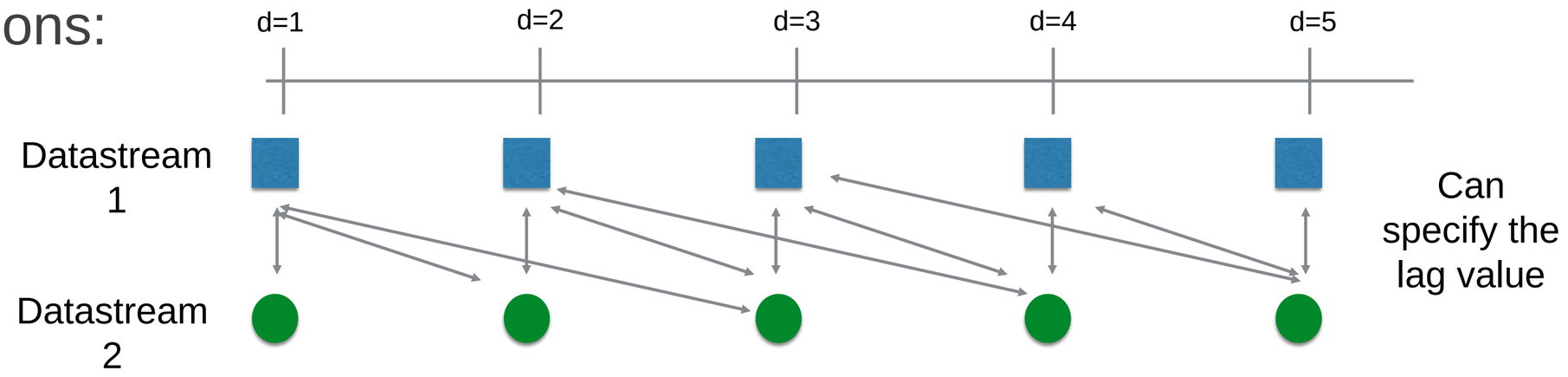
RMSE for Deciduous Needleleaf Forest

RMSE for Deciduous Broadleaf Forest

. . .

LVT: Additional capabilities

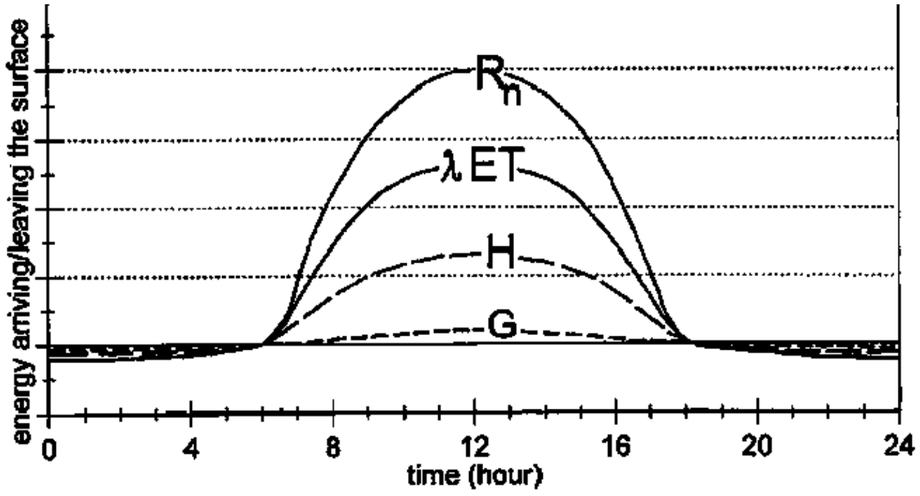
- Analysis outputs provided in both gridded (NetCDF/binary) and ASCII formats
- Time-lagged computations:



- Supports water-year (flexible year specification); user specifies the starting month of the year specification
- Smoothing support (limited); sees a moving window average for the computation of analysis metrics
- Computes confidence intervals (currently CIs in the spatial domain is supported; will be extended to include temporal CIs)

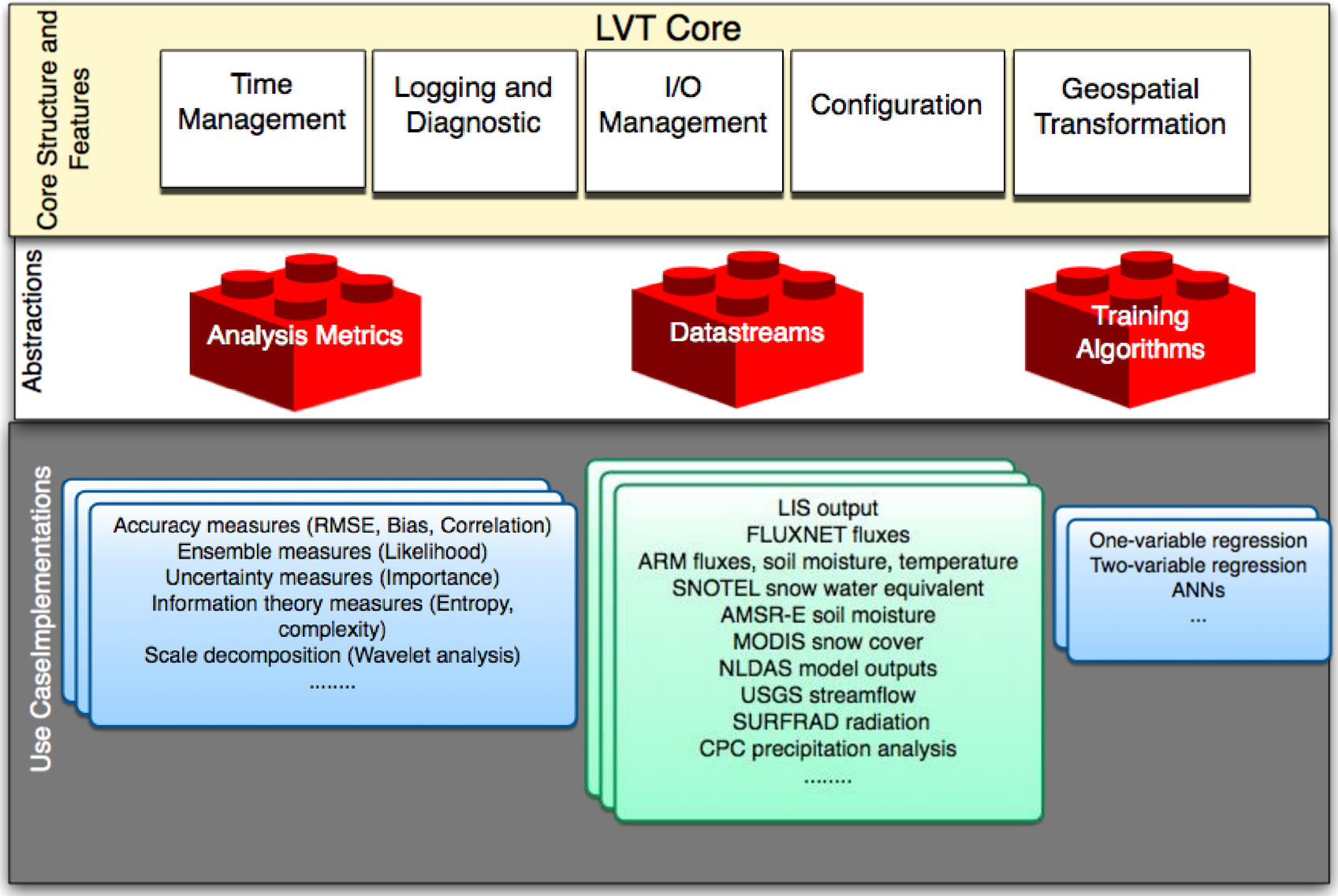
LVT: Additional capabilities

- Spatial-averaging modes for the analysis metric can be computed on a pixel-by-pixel basis or at basin-averaged basis:
  - Pixel-by-pixel - each pixel in datastream 1 is compared to a pixel in datastream 2
  - Basin-averaged - datastream 1 and 2 values are averaged to the basin scale and then compared using the analysis metric
- Computes derived variables:
  - e.g., Bowen ratio can be computed through LVT (and used for analysis) if both Qle and Qh are present; A column-averaged, weighted root zone soil moisture if individual soil moisture layer values are present
  - Energy/Water/Evaporation balance values
- Analysis metric computations are performed :
  - Across the entire analysis period
  - At specified temporal intervals
  - Average seasonal/diurnal cycles (if specified by the user)
- Supports outputs from all LIS computational subsystems:
  - Data assimilation diagnostics from the LIS-DA output

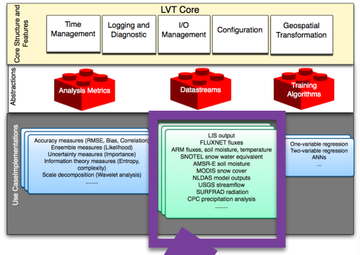


LVT: Software architecture

- 3-layer architecture
- An object-oriented framework with plugins defined for:
  - Analysis metrics
  - Datastreams
  - Training algorithms
- Analysis instances are enabled by a config file (no external scripting required)



LVT: Supported data streams



In-situ

- Ameriflux fluxes
- ARM fluxes, soil moisture, soil temperature
- ARS soil moisture
- CEOP fluxes, soil moisture, soil temperature
- CPC precipitation
- FLUXNET fluxes
- FMI SWE
- GHCN snow depth
- GLERL lake fluxes, temperature
- ISMN soil moisture
- NASMD soil moisture
- PBOH2O soil moisture, snow depth
- SCAN soil moisture
- SMOSREX soil moisture
- SNODEP snow depth metobs
- SNOTEL SWE
- SURFRAD radiation
- USGS streamflow
- USGS groundwater
- ..

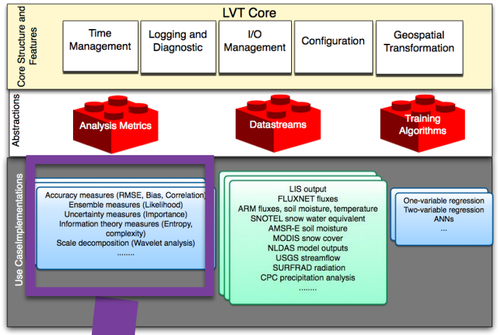
Satellite/Remote Sensing

- ALEXI
- AMSR-E SWE/snowdepth
- LPRM AMSR-E soil moisture
- ESA CCI soil moisture
- GIMMS NDVI
- GlobSnow SWE
- GRACE TWS
- ISCCP LST
- MOD10A1 snow cover
- MOD16A2 ET
- MODIS LST
- SMOPS soil moisture
- SMOS L1 Tb
- SMOS L2 soil moisture
- UW ET
- …

Model/Reanalysis

- AGRMET
- GLDAS2
- NLDAS2
- LIS outputs
- MERRA2
- SNODAS
- CMC
- GL6 JULES
- ERA interim Land
- MERRA Land
- COAMPS
- ….

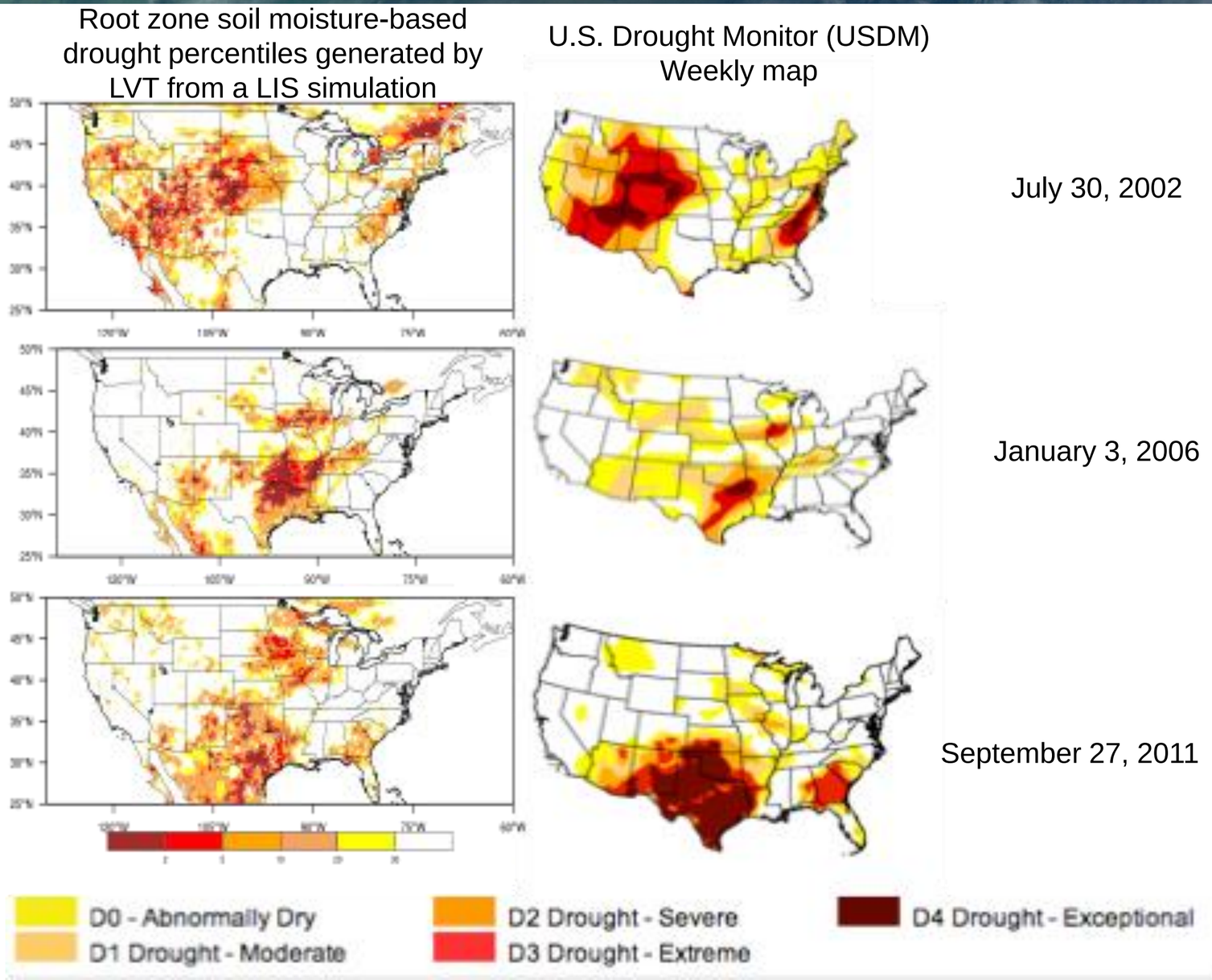
LVT: Supported analysis metrics



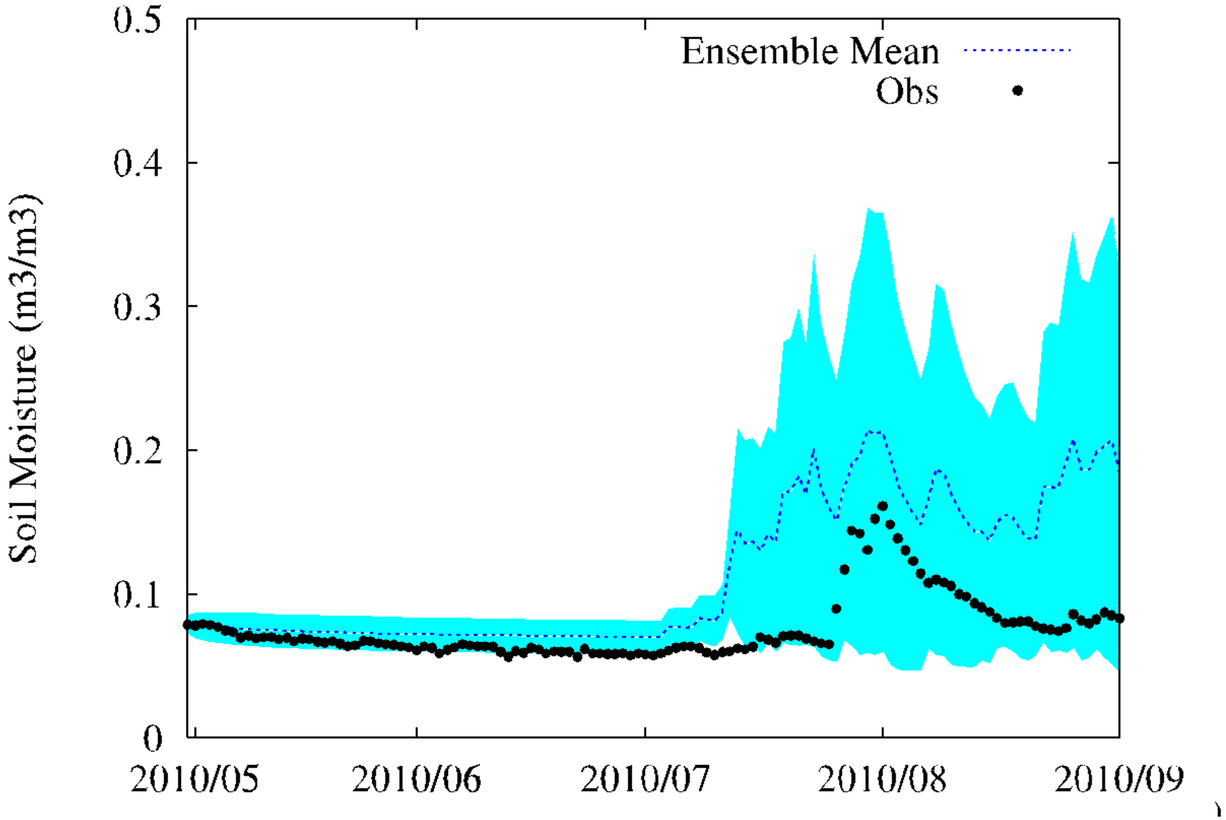
Metric class	Examples
Diagnostics	Mean, Standard deviation, Anomaly, Tendency, Min, Max, Sum, Maxtime, Mintime
Accuracy	ACC, Bias, CSI, ETS, FAR, FBIAS, MAE, NSE, PODY, PODN, POFD, Correlation, Anomaly Correlation, Tendency Correlation, unbiased RMSE
Indicators	SPI, SRI, SSWI, SSGI, percentiles, probabilistic percentiles
Ensemble	Mean, Likelihood, Spread, Cross correlation, ME
Information theory	Metric entropy, Information gain, Effective complexity, Fluctuation complexity
Scale decomposition	Discrete wavelet transforms
Spatial similarity	Hausdorff norm

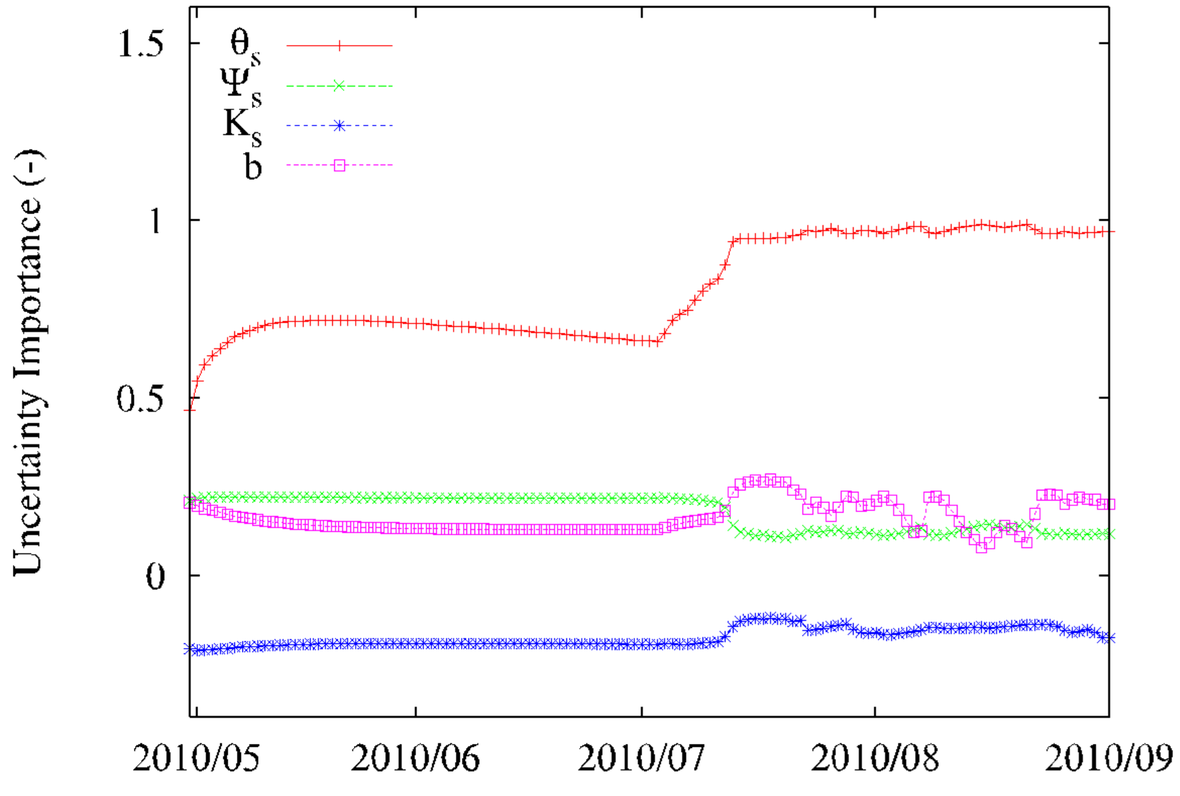
Example: Indicators from LVT

- LVT includes a suite of common, normalized indicators
- SPI, SRI, SSWI, SSGI, percentiles
- These indicators are computed as deviations from long term (fitted/computed) distributions



Example: LVT ensemble analysis





- Uncertainty importance: An assessment of the relative contribution of each parameter to the ensemble spread (cross correlation between the simulated variable and the parameter, across the ensemble)
- Can be used to guide parameter optimization/uncertainty estimation studies

LVT configuration: datastream attributes

```
LVT datastream attributes table::
SoilMoist  1 1  m3/m3   - 1 4 SoilMoist  1 1  m3/m3   - 1 1
::
```

- Variable name
- Starting number of selected levels (0=>not selected)
- Ending number of selected levels
- Units
- Direction type
- Time averaging option (0-instantaneous, 1- time averaged)
- Number of total vertical levels

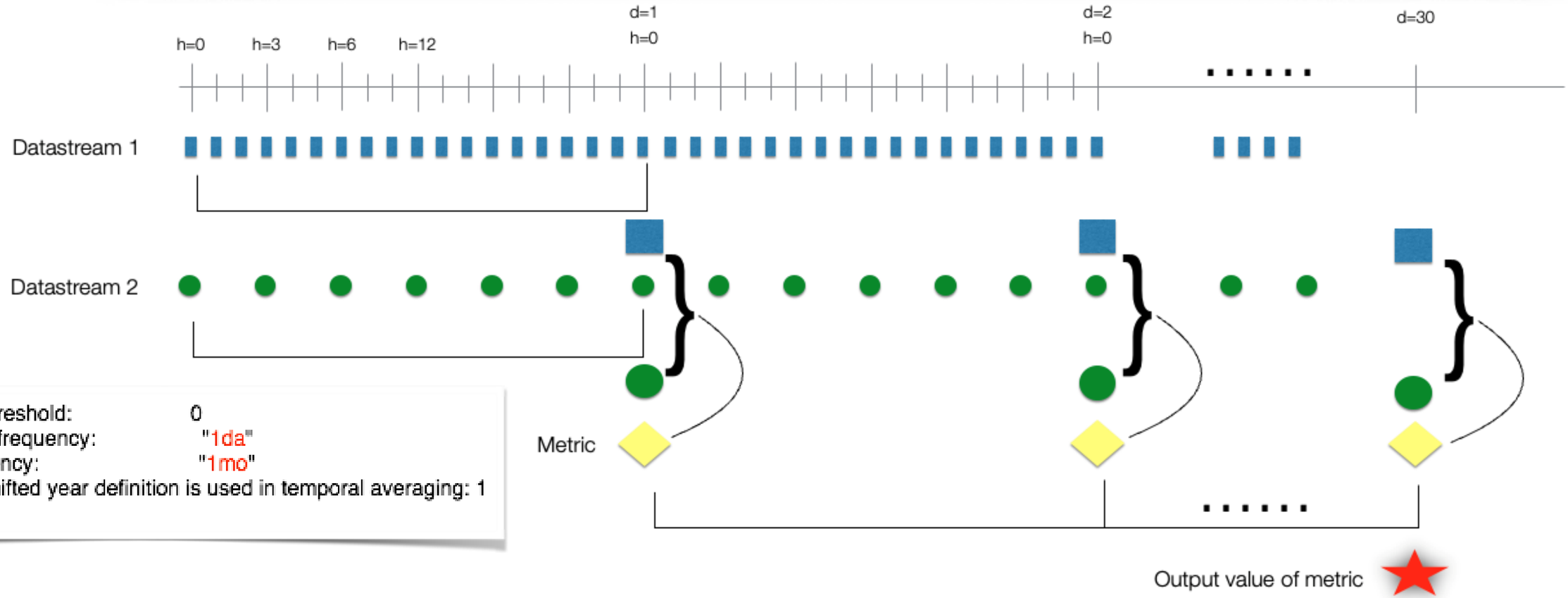
- Specifies the variables being analyzed from the datastreams
- Each line represents variable specification from datastream 1 and datastream 2
- Specification for each variable consists of 6 columns; variable names follow ALMA convention
- Any variable from datastream 1 can be compared to any variable from datastream 2 (as long as the metric of comparison is meaningful!)

```
LVT datastream attributes table::
Qle  1 1 W/m2   UP 1 1 SoilMoist   1 1 m3/m3 - 1 4
Qle  1 1 W/m2   UP 1 1 TotalPrecip 1 1 kg/m2 - 1 1
Qle  1 1 W/m2   UP 1 1 Rnet       1 1 W/m2  - 1 1
RadT 0 0 K     -  1 1 RadT       0 0 K     -  1 1
::
```

- Qle being compared against surface soil moisture, total precipitation, Net radiation
- Qle from datastream 1 is used for multiple comparisons against variables from datastream 2
- Radiative temperature comparison is turned off

LVT configuration: Output frequency

- Observation count threshold - computations are excluded over those grid points where the specified minimum count is not met
- Metrics computation frequency - the individual datastream values are averaged upto this interval and then the metric is computed
- Metrics output frequency - the metric values are averaged upto the stats output interval




```
Observation count threshold:                    0
Metrics computation frequency:                  "1da"
Metrics output frequency:                       "1mo"
Starting month if a shifted year definition is used in temporal averaging: 1
```

LVT configuration: Time series output

```
Time series location file:                    ./TS_LOCATIONS.TXT
```

- Specifies the name of the file that lists the locations and regions in the domain where ASCII time series data (for each metric) are to be derived
- Five different styles of specifying the locations/regions

Style 1: Specify lat/lon bounding boxes

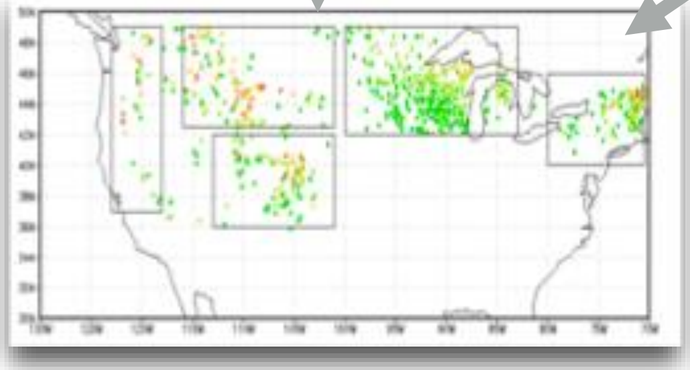
```
#Number of locations
2
#Location style (1-lat/lon, 2-col/row, 3-tile)
1
#Location name, (next line) SW-lat, SW-lon, NE-lat, NE-lon,
min number of grid points
WEST_US
40 -130 50 -110 5
HIGH_PLAINS_US
43 -110 49 -100 2
 .....
 .....
```

Style 2: Specify column/row boxes

```
#Number of locations
1
#Location style (1-lat/lon, 2-col/row, 3-tile)
2
#Location name, (next line) SW-col, SW-row, NE-col, NE-row,
min number of grid points
WEST_US
1  1  20 30 5
EAST_US
1  1  10 10 5
 .....
 .....
```

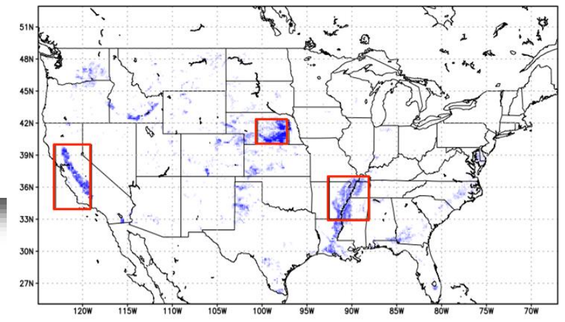
Style 3: Specify 1-d tile bounding indices

```
#Number of locations
1
#Location style (1-lat/lon, 2-col/row, 3-tile)
3
#Location name, (next line) Start index, Ending index,
min number of grid points
WEST_US
1  20 5
EAST_US
1  10 5
.....
.....
```



Style 4: List of lat/lons to specify a region

```
#Number of locations
2
#Location style
4
#number of points followed by lat/lon of each point
REGION1
3
34.4 -103.2
33.4 -100.2
32.1 -99.3
REGION2
2
40.2 -103.3
42.2 -104.2
```



Style 5: Regions defined by a categorical map

```
#Number of stations
3
#style
5
#names
NEWENGLAND
1
MIDATLANTIC
1
SOUTHATLANTIC
1
#categorical map
../huc02_conus_0.125dg.1gd4r
```



LVT: Journal article and documentation

LVT reference with a more complete description of the given components and features, please see:

Kumar, S.V., C.D. Peters-Lidard, J. Santanello, K. Harrison, Y. Liu, and M. Shaw, 2012: Land surface Verification Toolkit (LVT) - a generalized framework for land surface model evaluation. *Geosci. Model Dev.*, 5, 869-886, doi:10.5194/gmd-5-869-a

https://doi.org/10.5194/gmd-5-869-2012

Also, visit our LDT webpage (located within the LIS website):

https://lis.gsfc.nasa.gov/lvt-documentation